



Module Coursework Coversheet

Blind Grading Number: 8864G
Module Code: 4F13
Module Name: Probabilistic Machine Learning
Coursework Number or Name: 3: Latent Dirichlet Allocation
Word or Page Count (if limit prescribed): 5 pages (cover sheet excluded)

Declaration

☒ <i>tick to confirm</i>	I hereby declare that this piece of work is my own unaided effort and adheres to the Department of Engineering's guidelines on plagiarism.
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This piece of work is to be marked according to the following grading scheme:

Distinction		Pass				Fail (C+ - marginal fail)		
Outstanding	A+	A	A-	B+	B	C+	C	Unsatisfactory
90-100	80-89	75-79	70-74	65-69	60-64	55-59	50-54	0-49
Late Penalty: 10% of mark for each day, or part day, late (Sunday excluded).								

Coursework 3: Latent Dirichlet Allocation

Question A

Command 1 MLE Calculating

```
# A has rows (docID, wordID, count)
word_ids, word_counts = A[:, 1].astype(int) - 1, A[:, 2]
V_count = np.zeros(np.max(word_ids) + 1)
for wid, cnt in zip(word_ids, word_counts):
    V_count[wid] += cnt
# p(w) = count(w) / total_count
theta_ML = V_count / V_count.sum()
```

Figure 1 Top 20 words by MLE probability

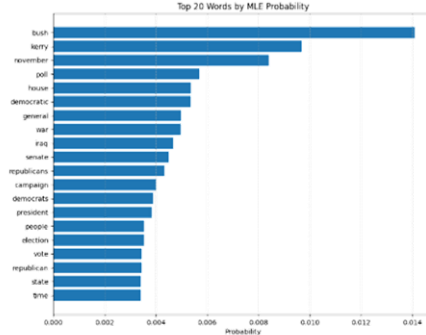


Figure 1 shows the 20 words with largest $\hat{\theta}_w$, mainly topical political words and common stop-words.

Maximum likelihood: We use a unigram (bag-of-words) model where each token is drawn i.i.d. from a categorical distribution with parameter $\theta = (\theta_1, \dots, \theta_M)$, $\sum_w \theta_w = 1$. Let c_w be the total count of word w in the training set and $N = \sum_w c_w$. Under the multinomial model as follows, maximizing $\log L(\theta)$ subject to $\sum_w \theta_w = 1$ gives $\hat{\theta}_w^{ML} = \frac{c_w}{N}$, which is what **Command 1** computes.

$$L(\theta) \propto \prod_w \theta_w^{c_w}, \log L(\theta) = \sum_w c_w \log \theta_w$$

Test-set log probability: For a test document with word counts c_w and total length $N_{\text{test}} = \sum_w c_w$, the log probability is $\log p(\text{doc} | \hat{\theta}) = \sum_w c_w \log \hat{\theta}_w$.

Maximum log probability: Let $\hat{\theta}_{\max} = \max_w \hat{\theta}_w$ be the largest MLE word probability. For fixed N_{test} , $\log p(\text{doc} | \hat{\theta})$ is maximized by putting all tokens on this single most probable word, so $\log p_{\max} = N_{\text{test}} \log \hat{\theta}_{\max}$.

Minimum possible test-set log probability: If the test vocabulary can contain any word type not seen in training, its MLE probability is $\hat{\theta}_{w_{\text{new}}} = 0$. Any test document that includes such a word has $p(\text{doc} | \hat{\theta}) = 0$, $\log p_{\min} = \log 0 = -\infty$. So the minimum possible test-set log probability for this MLE model is $-\infty$.

This illustrates the zero-probability problem of a pure MLE unigram. Any document containing an unseen word gets log-probability $-\infty$, while a very unnatural document that simply repeats the single most frequent word can achieve the highest possible log-probability. This makes evaluation extremely sensitive to vocabulary mismatch between train and test sets and motivates the use of smoothing or Bayesian methods in later questions.

Question B

We place a symmetric Dirichlet prior with concentration parameter α on the unigram probabilities $\theta = (\theta_1, \dots, \theta_W)$, $\theta \sim \text{Dir}(\alpha, \dots, \alpha)$, where W is the vocabulary size. With training counts N_w and total $N = \sum_w N_w$, the MLE from Question A is $p_{\text{ML}}(w | D) = N_w / N$. Under the Dirichlet prior the

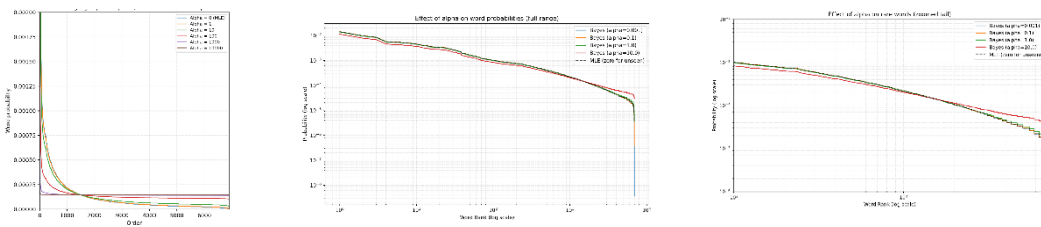
posterior is $\theta \mid D \sim \text{Dir}(N_1 + \alpha, \dots, N_W + \alpha)$, giving Bayesian predictive probabilities as follows, which is equivalent to adding a pseudo-count α to every word.

$$p_{\text{Bayes}}(w \mid D, \alpha) = \frac{N_w + \alpha}{N + W\alpha}$$

This can be written as $p_{\text{Bayes}}(w \mid D, \alpha) = \frac{N}{N + W\alpha} p_{\text{ML}}(w \mid D) + \frac{W\alpha}{N + W\alpha} \frac{1}{W}$, so the Bayesian predictive is a convex combination of the MLE and the uniform distribution $1/W$; in the limits $p_{\text{Bayes}} \rightarrow p_{\text{ML}}$ as $\alpha \rightarrow 0$ and $p_{\text{Bayes}} \rightarrow 1/W$ as $\alpha \rightarrow \infty$. Using the same counts as in Question A, we compute the Bayesian predictive probabilities for several values of α with **Command 2** and compare them with the MLE in **Figure 2**, using linear and log-log plots and a zoom into the tail.

Command 2 Bayesian unigram with Dirichlet(α) prior

```
theta_bayes = (word_counts + alpha) / (np.sum(word_counts) + W * alpha)
```



(a) Results on linear axes (b) Results on log-log axes (c) Results zoomed into the tail

Figure 2 Effect of the Dirichlet prior parameter α on unigram word probabilities.

For **high-frequency words** with large N_w , especially when α is small, the pseudo-count α is negligible, so $p_{\text{Bayes}}(w \mid D, \alpha) \approx p_{\text{ML}}(w \mid D)$ and the curves in **Figure 2** almost coincide in the head of the distribution, meaning the prior has little effect on very common words. For **rare or unseen words** with small N_w , the difference is large: under the MLE a word with $N_w = 0$ has $p_{\text{ML}}(w_{\text{new}} \mid D) = 0$, giving $\log p = -\infty$ for any test document containing it, whereas under the Bayesian model $p_{\text{Bayes}}(w_{\text{new}} \mid D, \alpha) = \alpha/(N + W\alpha) > 0$, which lifts the tail and avoids zero probabilities. Increasing α strengthens the prior: probabilities of common words are shrunk towards $1/W$ and probabilities of rare words are boosted, so the distribution becomes more uniform; when α is small the distribution is close to the MLE but with non-zero mass on unseen words. Thus the Dirichlet(α) Bayesian unigram smoothly interpolates between an unsmoothed MLE model and a strongly regularised almost-uniform model while resolving the zero-probability issue for rare and unseen words.

Question C

Using the Bayesian predictive distribution from Question B with $\alpha = 0.1$, the log probability of the test document with ID 2001 is $\log p(d_{2001} \mid D, \alpha) = \sum_w c_{w,2001} \log p_{\text{Bayes}}(w \mid D, \alpha) = -3691.22$, see the Categorical LP entry for $\alpha = 0.10$ in **Table 1**. We use the **categorical** likelihood rather than the multinomial as the multinomial adds a combinatorial factor $N_d! / \prod_w c_{dw}!$ that depends only on the document length N_d and therefore shifts all log-likelihoods for that document by the same constant, which is unhelpful when defining per-word perplexity.

Command 3 Compute Log Probability (Categorical) & Perplexity

```
lp_2001 = np.sum(counts_2001 * np.log(theta_bayes[indices_2001]))
perplexity_2001 = np.exp(-lp_2001 / np.sum(counts_2001))
```

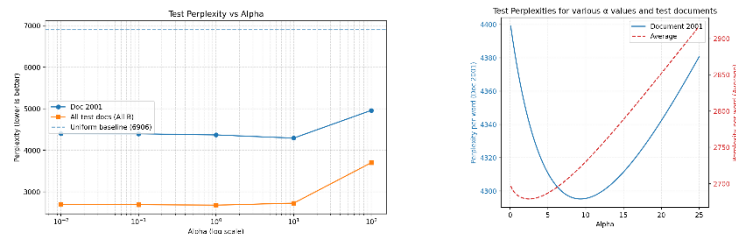


Figure 3 Test per-word perplexity of document 2001 and the full test set as a function of α

Per-word perplexity is $\text{ppl}(d) = \exp\left(-\frac{1}{N_d} \log p(d)\right)$, so it reflects how well the model predicts the specific word distribution of document d . Documents whose empirical word frequencies are close to the training distribution get lower perplexity, whereas documents like 2001 that contain more rare or idiosyncratic words get higher perplexity. **Table 1** and **Figure 3** show how $\text{ppl}(d_{2001})$ and $\text{ppl}(B)$ vary with α . Both curves are lowest for moderate values of α around 0.1–1. When α is very small the amount of smoothing is insufficient, the model assigns too little probability to rare words, and the perplexity increases. When α is very large the prior pushes the predictive distribution towards being almost uniform, the model underfits the data, and the perplexity also increases. From **Table 1**, the per-word perplexity for document 2001 is 4398.98 and the average per-word perplexity over all documents in B is 2697.11.

Table 1 Log Probabilities and Perplexity for varying α

Alpha (α)	Categorical LP	Multinomial LP	Perplexity (Doc 2001)	Perplexity (All B)
0.01	-3691.51	-1691.07	4401.93	2704.25
0.10	-3691.22	-1690.77	4398.98	2697.11
1.00	-3688.62	-1688.18	4373.11	2683.98
10.00	-3680.75	-1680.30	4295.55	2730.41
100.00	-3744.23	-1743.78	4962.19	3703.39

For a **uniform multinomial**, $\theta_w = 1/W$ for all w . Then every token has log-probability $\log(1/W) = -\log W$, and for any document or corpus we have:

$$\text{ppl}_{\text{uniform}} = \exp\left(-\frac{1}{N_{\text{test}}} \sum_{d,w} c_{dw} \log \frac{1}{W}\right) = \exp(\log W) = W$$

With vocabulary size $W = 6906$, the per-word perplexity of a uniform multinomial is 6906, higher than all Bayesian perplexities in **Table 1** and **Figure 3**, showing that the Bayesian unigram model captures meaningful structure in the data compared with a chance-level uniform model.

Question D

To monitor convergence I modified the provided `bmm.py` to record the posterior mixing proportions after each Gibbs sweep. With D documents, $K = 20$ components and a symmetric Dirichlet prior

$\pi \sim \text{Dir}(\alpha, \dots, \alpha)$, the posterior mean of the mixing weight for component k after sweep t is as follows, where $N_k^{(t)}$ is the number of documents currently assigned to component k .

$$\hat{\pi}_k^{(t)} = \mathbb{E}[\pi_k | z^{(t)}] = \frac{N_k^{(t)} + \alpha}{D + K\alpha},$$

Command 4 *Mixing proportions inside Gibbs loop*

```
props = (sk_docs.flatten() + alpha) / (D + K * alpha) # posterior mean  $\hat{\pi}$ 
history[t, :] = props
```

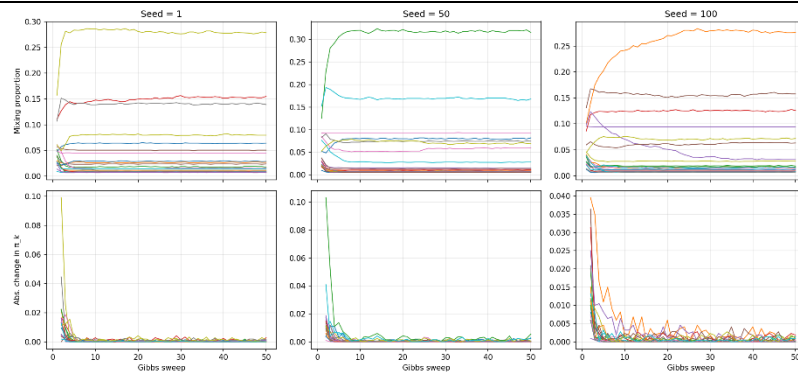


Figure 4 *Gibbs sampling mixing proportions and convergence diagnostics*

We run the sampler for 50 sweeps with three different random seeds (1, 50, 100). For each run we also record the step-to-step change $\Delta^{(t)} = |\hat{\pi}^{(t)} - \hat{\pi}^{(t-1)}|$. The top row of **Figure 4** shows the trajectories of $\hat{\pi}_k^{(t)}$ for each seed; the bottom row shows the corresponding $\Delta^{(t)}$. The resulting test-set perplexities for seed 1, 50 and 100 are: **2092.27**, **2117.59** and **2100.74**.

Across all three seeds the mixing proportions display a short **burn-in phase** (roughly sweeps 1–10), during which several components rapidly gain most of the mass and many others shrink towards zero. After this period the curves become flat and only fluctuate around stable means, indicating an approximately **stationary phase**. The diagnostic curves $\Delta^{(t)}$ drop sharply in the first few sweeps and then stay close to zero, showing that successive sweeps make only very small updates once burn-in has finished. Different seeds produce similar stable patterns of high- and low-weight components and very similar test perplexities, suggesting that the chains are sampling from comparable regions of the posterior. Within the scope of this coursework, these trace plots and diagnostics indicate that the Gibbs sampler has effectively converged to the posterior mixing proportions after about 10–20 sweeps, and that 50 sweeps are adequate for this mixture-of-multinomials model on the Daily Kos data.

Question E

Command 5 *Compute topic posteriors & word entropies per sweep*

```
topic_posterior[it, :] = (sk + alpha) / np.sum(sk + alpha)
pw = (swk + gamma) / (sk[None, :] + W * gamma) # p(w | k)
topic_entropy[it, :] = -np.sum(pw * np.log2(pw + 1e-12), axis=0)
```

To analyze convergence of the LDA model, we modified the code so that after each Gibbs sweep we record both the posterior topic proportions and the entropy of each topic's word distribution. Let sk_d be the number of tokens in document d assigned to each topic and sk the total token count for each topic across the corpus. With symmetric Dirichlet priors $\theta_d \sim \text{Dir}(\alpha, \dots, \alpha)$ and $\beta_k \sim \text{Dir}(\gamma, \dots, \gamma)$, the posterior expectations are as follows, where N_{wk} is the number of times word w is assigned to topic k and $N_k = \sum_w N_{wk}$:

$$\hat{\theta}_{dk} = \mathbb{E}[\theta_{dk} | z_d] = \frac{sk_{dk} + \alpha}{\sum_j (sk_{dj} + \alpha)}, p(w | k) = \frac{N_{wk} + \gamma}{N_k + W\gamma},$$

In the implementation I track the global topic proportions $\hat{\pi}_k = \frac{sk_k + \alpha}{\sum_j (sk_j + \alpha)}$ and the word entropy of each topic $H_k = - \sum_w p(w | k) \log_2 p(w | k)$, measured in bits as we use $\log_2$.

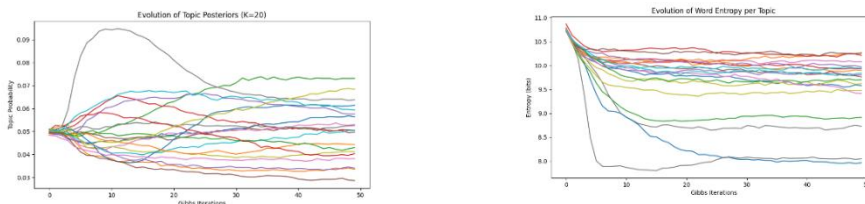


Figure 5 topic posteriors & entropies

Figure 5 (left) plots $\hat{\pi}_k$ over 50 sweeps. All topics start with similar probabilities; during the first 10–15 sweeps there is a clear burn-in phase where a few topics rapidly gain most of the mass and many shrink towards near zero. After roughly 20 sweeps the curves flatten and only show small random fluctuations, indicating that the sampler has reached an approximately stationary regime. Using the state after 50 sweeps I compute the per-word perplexity on the held-out test set B ; the LDA perplexity ($\approx 1.6 \times 10^3$) is much lower than for the Bayesian unigram in Question C ($\approx 2.7 \times 10^3$) and the mixture-of-multinomials model in Question D ($\approx 2.1 \times 10^3$), so 50 sweeps already give a model that fits the data substantially better than the earlier baselines and appears adequate for this dataset.

Figure 5 (right) shows the topic entropies H_k as a function of sweep. Initially, when assignments are almost random, all topics have high entropy, corresponding to diffuse, almost uniform word distributions. During the first 20–30 sweeps the entropies of most topics drop and then stabilise: low-entropy topics concentrate probability on a small, coherent set of words, whereas higher-entropy topics are broader or mix several subthemes. The fact that these entropy curves stabilise at different levels, in bits, indicates that topics have learned different degrees of specificity rather than remaining indistinguishable.

The final topic posteriors and entropy traces show neither a large group of almost-empty topics nor a set of 20 nearly identical, high-entropy topics. Instead, a modest number of topics carry substantial posterior mass and cover a range of entropy levels, while all topics are used to some extent. Together with the low test-set perplexity compared to simpler models, this suggests that $K = 20$ provides a **reasonable level of complexity** for the Daily Kos corpus. It is large enough to capture multiple political themes, but not so large that many components are redundant.